

customers. Productivity level decreases and legal liability increases. A stolen IP address alone could be fatal for a small high-tech company. To address network security, many small businesses have mitigated their risks by relying on the basics of marginal security with WEP in off-the-shelf solutions. Due to well-known cryptographic weaknesses, WEP is no longer a viable option. Today, WPA 802.1X/EAP is regarded as the most secure alternative for securing Wi-Fi connections.

What's A Small Business To Do?

WEP is not an adequate solution for even the smallest businesses like accounting, attorneys-at-law, and healthcare firms that handle sensitive client and financial data. Even a VPN solution is not foolproof. Although it strengthens WEP security, it does not protect the wireless link. Nor is it easy to configure. The current working WLAN security standard, WPA, requires choosing 802.1X authentication. Although the preferred mode of security, not every business—particularly small businesses—are capable of deploying the RADIUS server necessary to track users and their log-ins. For the small enterprise, this solution may be more complex and costly than necessary.

So, what's the alternative?

The alternative to a RADIUS solution is a WPA upgrade in combination with PSK—using PSK in place of 802.1X. However, the problems with PSK involve:

- o Scalability and Manageability
- o difficulties adding/deleting users
- o no support for guest access
- o does not provide individual user authentication so if values chosen are obvious; a user is left open to a dictionary attack.

Various security researchers fault PSK for the WPA security snag. Though WPA remains cryptographically secure, the PSK method makes it susceptible to passive monitoring and diction-